

EMT Disconnected One-Point Building Blocks

Flowed Quark/Gluon Loops, HDF5 Metadata, and Hierarchical Probing

PyQUDA Measurement Notes

May 19, 2026

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1 Purpose

This note documents the hadron-independent one-point functions used to build disconnected contributions to pion and proton energy-momentum tensor (EMT) matrix elements. The corresponding source file is `pyquda_measurement_utils/Disconnected_1pt_EMT_vibe_develop.py`.

The one-point workflows do not form a hadron three-point function by themselves. Instead, they provide flowed quark and gluon loop observables that are combined with hadron two-point functions in ensemble analysis.

2 Flowed One-Point Observables

Let t_f denote the gradient-flow time. The flowed gluonic building block is schematically

$$L_{\mu\nu}^g(q, t_f) = \sum_{\mathbf{x}} e^{i\mathbf{q}\cdot\mathbf{x}} O_{\mu\nu}^g(\mathbf{x}, \tau; t_f), \quad (1)$$

where $O_{\mu\nu}^g$ is the chosen flowed gluonic EMT operator.

The flowed quark loop is estimated stochastically. For a noise vector ξ_r and solution $\eta_r = D^{-1}\xi_r$, the measured loop has the form

$$L_{\mu\nu}^q(q, \tau; t_f) = \frac{1}{N_r} \sum_{r=1}^{N_r} \sum_{\mathbf{x}} e^{i\mathbf{q}\cdot\mathbf{x}} \text{Tr}_{sc} \left[\xi_r^\dagger(x, t_f) \Gamma_{\mu\nu}[U(t_f), D] \eta_r(x, t_f) \right]. \quad (2)$$

The operator $\Gamma_{\mu\nu}$ includes the symmetrized covariant derivative structure used by the EMT kernel. The code stores only the upper triangle $\mu \leq \nu$, and downstream analysis reconstructs the lower triangle by symmetry when needed.

3 Disconnected Combination with Hadron Two-Point Functions

For a hadron two-point function $C_2^H(p_f, t)$, the disconnected three-point building block is

$$C_{3,\mu\nu}^{H,\text{disc}}(p_f, p_i; t, \tau; t_f) = \langle C_2^H(p_f, t) L_{\mu\nu}(q, \tau; t_f) \rangle_{\text{cfg}} - \langle C_2^H(p_f, t) \rangle_{\text{cfg}} \langle L_{\mu\nu}(q, \tau; t_f) \rangle_{\text{cfg}}, \quad (3)$$

with

$$q = p_f - p_i. \quad (4)$$

The associated ratio is typically

$$R_{\mu\nu}^{H,\text{disc}}(t, \tau; t_f) = \frac{C_{3,\mu\nu}^{H,\text{disc}}(p_f, p_i; t, \tau; t_f)}{C_2^H(p_f, t)}, \quad (5)$$

before applying kinematic projections, vacuum-subtracted gradient-flow matching, and the perturbative mixing coefficients required for the renormalized EMT.

The subtraction term is essential: the one-point loop is hadron independent and contains a vacuum expectation value. The subtraction should be done at the ensemble-analysis level using the same gauge-configuration sample as the hadron two-point function.

4 Ringed-Fermion Diagnostics

The quark workflow also writes χ -type flowed-fermion diagnostics. These outputs are useful for monitoring or reconstructing ringed-fermion normalization factors. The standard kinetic normalization is not the raw χ dataset by itself; it should be reconstructed from the appropriate flowed kinetic expectation values, especially the zero-momentum diagonal EMT components.

5 Hierarchical Probing

The quark loop supports two source schemes:

$$\text{noise_scheme} \in \{\text{zn}, \text{hierarchical_probing}\}. \quad (6)$$

For ordinary Z_n noise, each stochastic source corresponds to one inversion. For hierarchical probing, a base noise vector ξ_b is multiplied by a set of site-only Rademacher vectors $h_k(x) = \pm 1$:

$$\xi_{b,k}(x) = h_k(x) \xi_b(x). \quad (7)$$

The effective number of inversions is therefore

$$N_{\text{eff}} = N_{\text{base}} N_{\text{HP}}. \quad (8)$$

In the application this is

$$\text{effective_n_inversions} = \text{EMT_1PT_N_VEC} \times \text{EMT_1PT_HP_NUM_VECTORS}. \quad (9)$$

EMT_1PT_HP_NUM_VECTORS must be a positive power of two.

5.1 Ordering Choices

Two Gray-code ordering choices are implemented:

Ordering	Meaning
global_xyzt_gray_projected_to_evenodd	Gray code the full x, y, z, t site id.
spatial_xyz_then_t_gray_projected_to_evenodd	Gray code spatial sites first, then append time bits.

The global ordering is the default to preserve the previously validated behavior. The spatial-first ordering is intended for studies where the leading noise correlations are expected to be dominated by spatial locality and where time dilution is not yet being used.

Spin-color dilution and time dilution are not part of the current short-term workflow.

6 HDF5 Output

The quark HDF5 file stores metadata attributes:

Attribute	Meaning
<code>flow_type</code> , <code>flow_epsilon</code> , <code>flow_steps</code> , <code>flow_times</code>	Gradient-flow schedule.
<code>qext</code> , <code>pf</code> , <code>p_2pt</code>	Momentum labels used by downstream analysis.
<code>volume_norm</code>	Spatial volume normalization used in the averaged data.
<code>upper_triangle_only</code>	Only $\mu \leq \nu$ components are written.
<code>noise_scheme</code> , <code>hp_num_vectors</code> , <code>hp_ordering</code>	Source-generation scheme.
<code>n_base_noise</code> , <code>effective_n_inversions</code>	Stochastic source counting.

The raw quark datasets are:

$$\text{raw/Tmunu_pervec}, \quad \text{raw/CHI_pervec}, \quad \text{raw/source_index}, \quad \text{raw/base_noise_index}, \quad \text{raw/hp_index}. \quad (10)$$

The bookkeeping datasets allow analysis code to group effective HP sources back to the original base noise:

Dataset	Meaning
<code>source_index</code>	Effective source index after HP expansion.
<code>base_noise_index</code>	Original stochastic base-noise index.
<code>hp_index</code>	Hierarchical-probing vector index for that base noise.

7 Current Validation Status

The current implementation has been checked with:

- Python syntax compilation.
- A CPU/numpy backend check that $\text{HP} = 1$ leaves the base source unchanged.
- A Perlmutter login-node GPU smoke test with `hp_num_vectors = 2` and spatial-first ordering.

Before production analysis, the recommended next validation is a scaling study comparing Z_n , $\text{HP} = 1$, $\text{HP} = 2, 4, 8, 16$ at fixed gauge, flow time, and momentum.